

Differentiation - Set 2

TMUA practice questions by topic. Specification MM6.1-MM6.3, MM8.5. 10 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.
- A 5
B $2\sqrt{2}$
C 4
D 0
E 2
F 8
- 2 Find the sum of the y-coordinates of the stationary points of the curve $y = 2x^3 - 9x^2 + 12x - 3$.
- A 2
B 0
C 5
D 4
E 3
F 1
- 3 Given $f(x) = x^{\frac{3}{2}} - 6x^{\frac{1}{2}}$ for $x > 0$, find the value of x at which f has its minimum.
- A 2
B 4
C 1
D 6
E $\sqrt{2}$
F 3

- 4 Which of the following statements about $f(x) = x^3 - 3x$ is/are true?
- I f is increasing for all $x > 1$.
 - II f has a local maximum at $x = -1$.
 - III $f(x) \geq -2$ for all real x .
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III
- 5 $f(x)$ is a polynomial with $f'(x) = (x - 2)^2(x + 1)$. Which of the following is true?
- A f has local minima at $x = -1$ and $x = 2$
 - B f has no turning points
 - C f has a local minimum at $x = -1$ and no turning point at $x = 2$
 - D f has a local maximum at $x = -1$ and a local minimum at $x = 2$
 - E f has a local minimum at $x = -1$ and a local maximum at $x = 2$
- 6 The volume of a cuboid with a square base of side x is $V = 4x^2 - x^3$ for $0 < x < 4$. Find the maximum volume.
- A 16
 - B $\frac{64}{27}$
 - C 8
 - D $\frac{32}{27}$
 - E $\frac{256}{27}$
 - F $\frac{128}{27}$

- 7 The tangent to $y = x^3$ at the point (a, a^3) , where $a \neq 0$, meets the curve again at the point with x -coordinate
- A $a/2$
 - B $-2a$
 - C $2a$
 - D 0
 - E $-a$
 - F $-a/2$
- 8 Which of the following statements is/are true for every function f that has a derivative at every real number?
- I If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.
 - II If f has a local minimum at $x = a$ then $f'(a) = 0$.
 - III If $f'(x) > 0$ for all x , then $f(3) > f(2)$.
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III
- 9 Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

10 The function f has $f'(x) = (x - 1)(x - 3)^2$. Which of the following statements is/are true?

- I** f has a local minimum at $x = 1$.
- II** f has a local maximum at $x = 3$.
- III** f is increasing for all $x > 1$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III